



Figure 3: MySQL screenshot



Figure 4: UserController screenshot

```
class User {
    long userid
    String username
    String password
    static constraints = {
        userid(unique:true)
        username(blank: false)
        password(blank:false)
    }
}
```

5. To create a controller for *User.groovy*, type:

```
grails create-controller test.User
```

Update the file `UserController.groovy` under `grails-app/controller/test`.

Inside the controller, we will define a scaffold that will automatically create the CRUD (Create, Read, Update, Delete) function for our domain class.

Your final *UserController.groovy* will look like what follows:

```
package test
class UserController {
    def scaffold = User
    def index() { }
}
```

Let's check everything out by running our application, so run the following in your terminal:

```
grails run-app
```

Now, if you check out your MySQL databases, you'll notice that a table called 'user' has been created. The *userid* field has a constraint of being unique, while the 'username' and 'password' fields are strings that cannot be blank.

If you check the contents of the table called `'user'`, user it'll be empty. You can check this with the following command.

```
select * from user;
```

Let's navigate to the `http://localhost:8080/test` URL.

There we will be able to see our new controller. Click on it to start CRUD activities.

Next, let's create a new user by clicking on the 'new user' button.

Finally, let's check if our data has been inserted into our table in MySQL.

This is how you integrate MySQL with a Grails application. Grails is not limited to MySQL. You can also use any other database like PostgreSQL or MongoDB.

Advice for the veterans

This section gives you some of the best practices in Grails, which will help you rapidly develop applications and tools with this framework.

1. *Always use a domain-driven design:* First, create your basic domain model classes and then use scaffolding to get them online. This will help you stay motivated and understand your domain better.
2. *Follow Grails conventions:* Grails is a convention-driven development environment.
3. *Use `DataSource.groovy`:* Use `DataSource.groovy` to configure data sources.
4. *Use `BuildConfig.groovy`:* This is to install any plugin in your



Figure 5: New controller added to Grails